

Module 3 — DADA2 in Galaxy to phyloseq

BIOL 4810

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Module 3 — DADA2 in Galaxy to phyloseq

Goal: Process amplicon sequencing reads in Galaxy using DADA2, examine key outputs, then prepare the files needed for analysis in **phyloseq**.

Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

0) Your dataset

0A. Sample set / project name:

WCSP Microbiome

0B. Sequencing type:

16S V4 Paired-End

0C. Metadata grouping variable you plan to analyze in phyloseq:

SampleSource

DADA2 workflow order

Use the Galaxy tools in this order:

```
plotQualityProfile → filterAndTrim → learnErrors → dada → mergePairs →  
makeSequenceTable → removeBimeraDenovo → assignTaxonomy → sequence  
counts
```

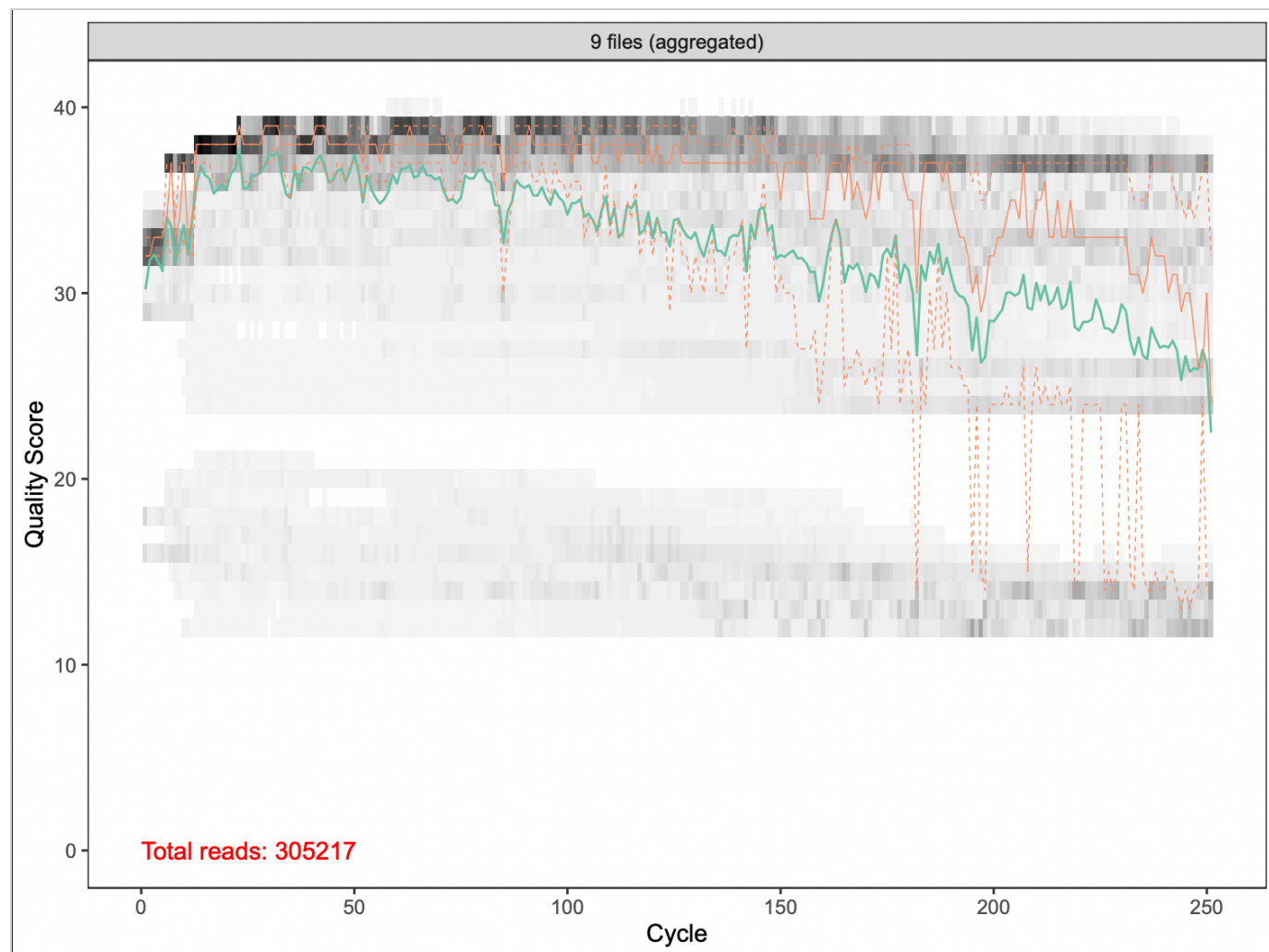
Reminder: The point of this workflow is to turn messy raw reads into a clean ASV table with taxonomy that can be analyzed ecologically in **phyloseq**.

Part 1 — Quality profiles

1A) Forward read quality profile

What is the purpose of this step?

To make sure that our sample is clear of any contaminants.

Upload screenshot: forward plotQualityProfile PDF Screen Shot... 9.46.50 AM

Where does quality begin to noticeably decline in the forward reads?

Chosen forward truncation length:

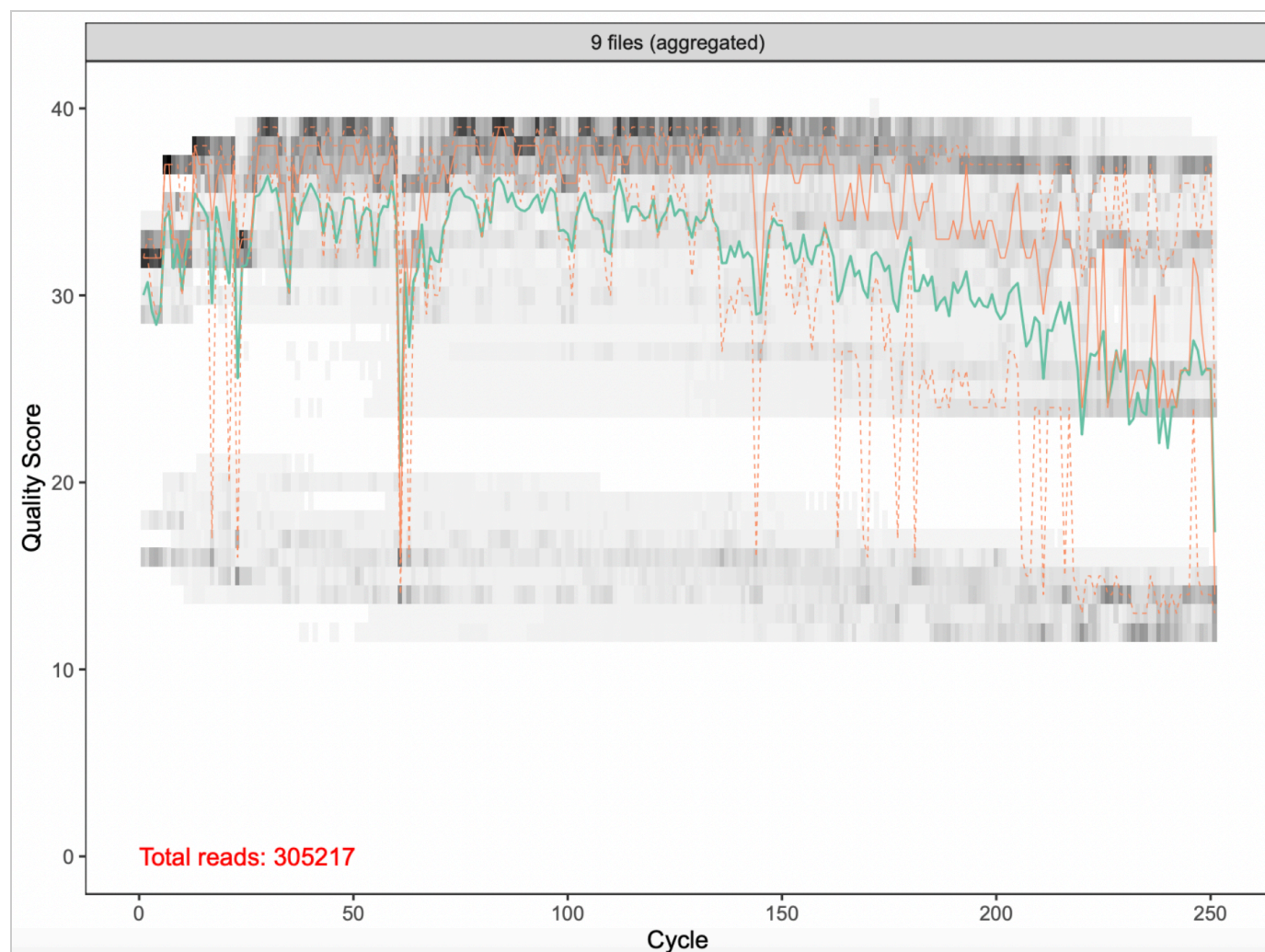
1B) Reverse read quality profile

What is usually different about reverse-read quality compared with forward-read quality?

Forward reads are also better than reverse reads since reverse reads are generated later, allowing for possible degradation.

Upload screenshot: reverse plotQualityProfile PDF

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Where does quality begin to noticeably decline in the reverse reads?

Chosen reverse truncation length:

1C) Truncation decision

Explain why you chose these truncation values.

A quality score above 30 is generally considered "good", so the value chosen are the ones who came below 30.

Rule of thumb: trim where quality begins to drop, but still keep enough overlap for forward and reverse reads to merge.

Part 2 — filterAndTrim

What filterAndTrim does:

This step removes low-quality reads and trims poor-quality bases from the ends of reads before denoising.

Why is this step important?

It cleans the data so that we're left with only high-quality pairs.

Parameters used:

Truncate read length: 220

Remove short reads: 20

Trim start of each read: 10

Truncate read length: 205

What could happen if you trim too aggressively?

You can get stuck with loss coverage, meaning you won't be able to see the whole picture.

What could happen if you do not trim enough?

You'll end up with the wrong information.

IMPORTANT — Split Paired Reads (Unzip Collection)

After `filterAndTrim`, your output is a **paired collection**.

Before continuing, you must split this into **forward and reverse collections**.

What to do in Galaxy

1. In the Tools panel, search: **unzip collection**
2. Input: your `filterAndTrim` paired output
3. Run

What you will get

- One collection = **forward reads (R1)**
- One collection = **reverse reads (R2)**

Why this matters

`learnErrors` and `dada` must be run **separately** on forward and reverse reads.
If you skip this, `mergePairs` will fail.

Part 3 — learnErrors

What learnErrors does:

This step models the sequencing error rates in your dataset so DADA2 can distinguish real sequence variants

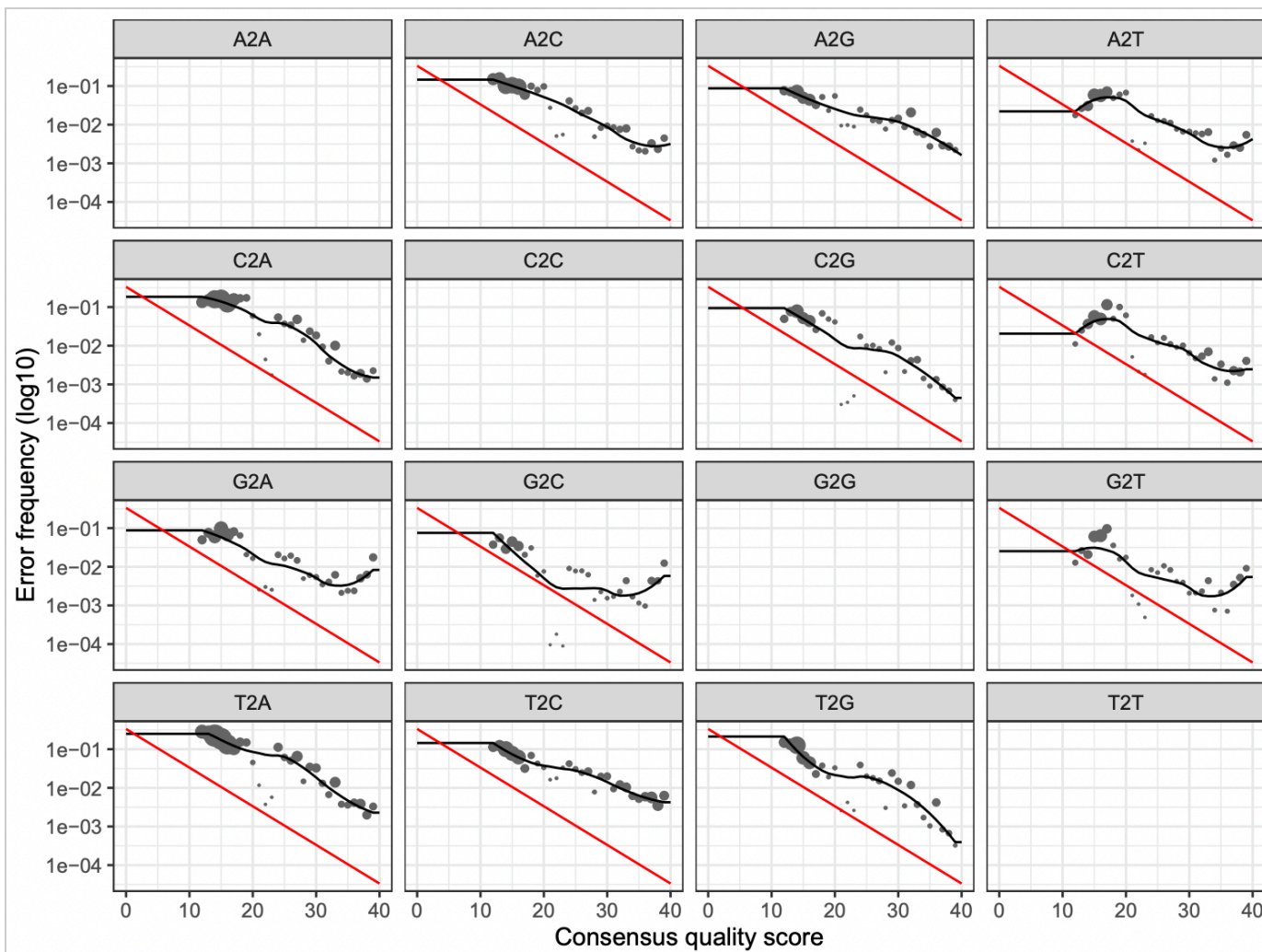
from sequencing mistakes.

Why is this step important?

We do this to remove reads that have too many errors in order to help us avoid the misidentification of species.

Upload screenshot: learnErrors PDF

Choose File Screen Shot...0.34.53 AM



Does the fitted model generally follow the observed error pattern?

Mostly Yes

Why is learning error rates important before denoising?

Learning error rates helps prevent you from removing too much info when you denoise.

Part 4 — dada (denoising)

What dada does:

This is the denoising step. DADA2 uses the learned error model to correct likely sequencing errors and infer exact amplicon sequence variants (ASVs).

What is denoising doing?

Denoising helps clean up the information by removing anything deemed unnecessary so that you have a clear picture of the information you're seeking.

How is ASV inference different from OTU clustering?

OTU is based on assumptions of what the program thinks the data means while ASV does more investigation in order to provide a more accurate answer.

Part 5 — mergePairs

What mergePairs does:

This step merges the forward and reverse reads for each amplicon so you can reconstruct the full sequence from overlapping read pairs.

What is being merged, and why?

The forward and reverse reads from each amplicon so that the full sequence can be reconstructed from overlapping read pairs.

Why do forward and reverse reads need to overlap?

So that the original DNA fragment can be recreated.

If reads fail to merge, what is one likely reason?

The reads were too long or too short.

Part 6 — makeSequenceTable

What makeSequenceTable does:

This step creates the ASV table. Each row is a sample, each column is an ASV, and each cell contains the number of times that ASV was observed in that sample.

What does this table represent biologically?

ASV tables show the amount of bacteria present and the types of bacteria present.

What are the rows, columns, and cell values in the sequence table?

The rows are the amplicon sequence variants (ASVs), columns are our samples, and the cell values are the counts.

Part 7 — removeBimeraDenovo**What removeBimeraDenovo does:**

This step removes chimeras, which are artificial sequences created during PCR when fragments from different templates are incorrectly joined together.

What is a chimera, and why should it be removed?

During PCR, fragments from different templates can sometimes be incorrectly joined together, resulting in an artificial sequence known as a chimera. These chimeras should be removed because they can be misleading

Why are chimeras a problem for interpretation?

As previously mentioned, chimeras can be misleading and lead to incorrect interpretations of results.

Part 8 — assignTaxonomy**What assignTaxonomy does:**

This step compares your ASVs to a reference database and assigns taxonomy, such as phylum, class, order, family, genus, or sometimes species.

What does taxonomy assignment add to your ASV table?

It allows us to know what groups our bacteria belongs to. Knowing the groups allows us to make assumptions of our bacteria.

Why might some ASVs not be assigned all the way to species?

We could be lacking the information needed in order to assign all the way.

Part 9 — Sequence counts summary**What sequence counts shows:**

This output tracks how many reads remain after each major step in the pipeline so you can see where reads were retained or lost.

Why is it useful to track reads across steps?

If something goes wrong, tracking the steps allows us to know exactly when and where the issue occurred.

Upload screenshot: sequence counts output

No file chosen

Approximately how many reads did you start with?

N/A

Approximately how many reads remained at the end?

N/A

At which step did you lose the most reads?

N/A

What is one likely explanation for that loss?

N/A

Part 10 — Moving into phyloseq

After completing DADA2 in Galaxy, we can create a phyloseq object for analysis.

10A. Which files will you take from DADA2 from Galaxy to use in phyloseq?

Metadata CSV, Bimera CSV, & Assign Taxonomy

10B. Why must sample names in the metadata match sample names in the ASV table?

The tools we use won't understand if there is a mismatch name.

10C. In 2–3 sentences, explain how DADA2 transforms raw reads into data that can be analyzed ecologically in phyloseq.

DADA2 does a lot repeated cleaning and organizing of the raw reads in order to transform it into analyzable data. It also includes a join stage where it combines the forward and reverse read in order to get a single complete read.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview**Dataset**

Project/sample set: WCSP Microbiome

Sequencing type: 16S V4 Paired-End

Grouping variable: SampleSource

Quality profiles

Purpose: To make sure that our sample is clear of any contaminants.

Reverse vs forward quality: Forward reads are also better than reverse reads since reverse reads are generated later, allowing for possible degradation.

Forward quality drop: 220

Reverse quality drop: 205

Forward truncation: 220

Reverse truncation: 205

Why these values: A quality score above 30 is generally considered "good", so the value chosen are the ones who came below 30.

filterAndTrim

Purpose: It cleans the data so that we're left with only high-quality pairs.

Parameters used:

Truncate reads at quality threshold: 2

Truncate read length: 220

Remove short reads: 20

Trim start of each read: 10

Truncate read length: 205

If trimmed too much: You can get stuck with loss coverage, meaning you won't be able to see the whole picture.

If not trimmed enough: You'll end up with the wrong information.

learnErrors

Purpose: We do this to remove reads that have too many errors in order to help us avoid the misidentification of species.

Model fit: Mostly Yes

Why important: Learning error rates helps prevent you from removing too much info when you denoise.

dada + mergePairs

Denoising purpose: Denoising helps clean up the information by removing anything deemed unnecessary so that you have a clear picture of the information you're seeking.

ASVs vs OTUs: OTU is based on assumptions of what the program thinks the data means while ASV does more investigation in order to provide a more accurate answer.

mergePairs purpose: The forward and reverse reads from each amplicon so that the full sequence can be reconstructed from overlapping read pairs.

Need for overlap: So that the original DNA fragment can be recreated.

One reason reads fail to merge: The reads were too long or too short.

Sequence table + chimera removal + taxonomy

Sequence table purpose: ASV tables show the amount of bacteria present and the types of bacteria present.

Rows/columns/cells: The rows are the amplicon sequence variants (ASVs), columns are our samples, and the cell values are the counts.

Chimera removal purpose: During PCR, fragments from different templates can sometimes be incorrectly joined together, resulting in an artificial sequence known as a chimera. These chimeras should be removed because they can be misleading and lead to incorrect interpretations of results.

Why chimeras are a problem: As previously mentioned, chimeras can be misleading and lead to incorrect interpretations of results.

Taxonomy purpose: It allows us to know what groups our bacteria belongs to. Knowing the groups allows us to make assumptions of our bacteria.

Taxonomy limits: We could be lacking the information needed in order to assign all the way.

Sequence counts

Purpose: If something goes wrong, tracking the steps allows us to know exactly when and where the issue occurred.

Starting reads: N/A

Ending reads: N/A

Largest loss step: N/A

Likely explanation: N/A

Move to phyloseq

Files to download:

Metadata CSV, Bimera CSV, & Assign Taxonomy

Why names must match: The tools we use won't understand if there is a mismatch name.

DADA2 to phyloseq transition: DADA2 does a lot repeated cleaning and organizing of the raw reads in order to transform it into analyzable data. It also includes a join stage where it combines the forward and reverse read in order to get a single complete read.